

GI Hub - Standard Operating Procedure

General Industries - GI Hub ERP

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General Industries Hub - Standard Operating Procedure

2026-07-18 note: the procedures below (roles, procurement chain, DN state machine, approval gates) apply UNCHANGED to the new React/FastAPI application - screens moved, the process did not. For per-screen instructions use the Product Manual (USER_MANUAL.md at the repo root; ops PDF in docs/export/GI-Hub-User-Manual.pdf). *(There was a second, role-based docs/USER_MANUAL.md from 2026-07-26; it fell three phases behind and was deleted in Phase 8 - there is one manual now.)* New since v1.0 and covered there: Surface-Shields system-first issuing, the SME Smart Calculator, Bulk Excel Import, handwritten-OCR validation + TSV export, and the admin Bug Tracking Engine.

Version 1.0 - Procurement Chain (Logistics + Warehouse + Site)

Owners: Logistics Manager - Warehouse Lead - Site HOD Council

Applies to: All sites + warehouses connected to the GI Hub ERP

Companion docs: USER_MANUAL.md (every page/tab/button) - handoff.md (technical architecture)

Cadence: Review quarterly; update on any role change, vendor onboarding, or DN-state-machine alteration.

§1. Purpose & Scope

This SOP turns the procurement-chain features of GI Hub v3.0 into a daily / weekly operating rhythm. It answers "what should I do at 8 AM Tuesday?" - for that, USER_MANUAL is too granular and handoff.md too technical.

In scope:

- Daily / weekly / monthly cadences for Site HOD, Logistics, Warehouse User, Store Keeper, Admin
- Decision trees for the four most common judgment calls (reject vs reschedule vs force-close, return vs adjustment, override timing, when to close)
- Escalation matrix - who-pings-who-after-how-long, with WhatsApp / email / admin override
- Recovery procedures - DN rejected at HOD, warehouse short-receives, vendor return + resupply, HOD absent
- Quick reference cards per role

Out of scope:

- UI screenshots (use USER_MANUAL.md)
- Code / SQL details (use handoff.md)
- Hosting + backup procedures (use handoff.md §4 and USER_MANUAL §17)

§2. Role Responsibility Matrix (RACI)

R = Responsible (does the work) - A = Accountable (signs off) - C = Consulted (input) - I = Informed (notified)

Activity	Site HOD	Logistics	Warehouse User	Store Keeper	Admin
Create PR	R/A	I	-	-	I
Submit PR to Logistics	R/A	I	-	-	I
Negotiate vendor + price	I	R/A	-	-	C
Issue PO	I	R/A	-	-	I
Upload PO PDF	-	R/A	-	-	-

Activity	Site HOD	Logistics	Warehouse User	Store Keeper	Admin
Assign PO to a warehouse	I	R/A	I	-	I
Acknowledge assignment	-	I	R/A	-	-
Receive goods from vendor	-	I	R/A	-	-
Prepare DN (RL/BL separated)	-	I	R/A	-	-
Approve DN delivery date	-	R/A	I	-	I
Approve DN content	R/A	I	I	-	I
Mark DN as received (site)	I	I	I	R/A	-
Request reschedule	C/R	A	C/R	-	I
Raise vendor return	C	A	C/R	C	I
Force-close PR / PO / line	I	R/A	-	-	I (critical)
Approve user registrations	-	-	-	-	R/A
Maintain vendor / warehouse master	-	C	C	-	R/A
Review audit logs	-	-	-	-	R/A
Resolve oversight escalations	C	C	C	C	R/A

§3. Cadence Checklists

§3.1 Site HOD - Daily

Morning (8:00 AM):

- [] Open HOD Portal -> check the bell badge in the sidebar
- [] Open inbox -> mark non-critical items read; act on critical (red border)
- [] EOD Commit tab - confirm yesterday's commits posted cleanly (no pending rows from yesterday)
- [] Pending Receipts - review and approve any inbound receipts (legacy path)
- [] DN Approvals (NEW) - approve or reject any DNs the Warehouse has prepared
- [] In-Transit -> "Active in-transit" sub-tab - scan ETAs for the day

Through the day:

- [] When PRs are raised - submit to Logistics within 1 hour (Submit PR(s) to Logistics expander)
- [] Reschedules from team - verify before submitting, include real reason
- [] EOD Commit by 17:00 - typed COMMIT confirmation

Evening (before sign-off):

- [] In-Transit -> confirm tomorrow's expected DNs are still on schedule

- [] Force-closures sub-tab - anything new since last login?
- [] Sign out cleanly (audit log)

§3.2 Site HOD - Weekly (Sunday or first workday)

- [] Purchase Requests - clean up any PR in site_draft longer than 7 days (either submit or delete)
- [] In-Transit -> My reschedule requests - chase any "pending" older than 48 hours
- [] Force-closures - review the past 7 days; raise an admin ticket on any without clear reason
- [] Cross-Site requests - fulfill outstanding incoming requests

§3.3 Site HOD - Monthly

- [] PR backlog review - anything still in submitted/in_po from > 30 days ago needs a call to Logistics
- [] Generate PO Status report scoped to your site - review on-time delivery rate
- [] Audit your own bell inbox for missed criticals (filter to unread)

§3.3a Site HOD - Man-Hours & Labor Tracking (Man-Hours)

Daily (or per shift):

- [] Daily Timesheet - log the day's labor. Either upload the day's attendance Excel (Replace rows for these dates so a re-upload never duplicates) or use the manual per-day batch grid (pick Work Date -> Equipment Tag -> System Code -> tick who worked + In/Out).
- [] Record team SQM completed for each tag/system worked, and pick the distribution (even / by-hours).
- [] Confirm Location/Equipment/System are filled on imported rows (the Excel arrives with them blank).

Weekly:

- [] Estimate vs Actual - review the variance table; for every red (>10 % over) scope, record a Reason for the over-consumption.
- [] Employees - deactivate workers who have left; confirm OWN vs Supply + Company are correct.
- [] Employee-wise - spot-check a few workers' date-wise timelines for gaps or double-bookings.

Monthly:

- [] Estimator - update required man-hours for upcoming scopes (and Estimated SQM for a MH-per-SQM norm).
- [] Compare total Actual man-hours vs Estimated for the month; flag systemic over-runs to management with the captured reasons.

Man-Hours is site-scoped and isolated - it never changes inventory or the EOD ledger. Equipment-Tag / System-Code drop-downs come from the Material Estimator master, so keep that seeded for the site.

§3.3b Site HOD - Material Estimator (Material Estimator)

- [] Before a work package: open Session/Selective Order -> pick equipment + SQM -> review the material requirement and the procurement total; raise PRs for shortfalls via the normal HOD PR flow.
- [] Weekly: Dashboard / Total Overview - check available-vs-required per location; the figures are computed live from receipts and committed consumption (no manual upkeep).
- [] Master Data: only Admin/HOD edits Equipment/Recipe/Materials here; these are SME-owned and never touch the ERP inventory ledger.

§3.4 Logistics - Hourly (during work hours)

- ☐ bell badge - clear unread within 30 minutes during business hours
- ☐ Incoming PRs queue - pick up new PRs within 4 hours; acknowledge to Site HOD via WhatsApp if delayed
- ☐ Reschedules - decide within 2 hours during business hours; out-of-hours, EOD next business day

§3.5 Logistics - Daily

Morning (8:00 AM):

- ☐ Run the day's PO PDF backlog (any vendor PDFs sitting in your email)
- ☐ Open POs -> filter to "Expected" = today + tomorrow -> confirm each has a warehouse assignment
- ☐ Open POs -> filter to "Expected" = yesterday -> if not delivered, chase warehouse + vendor
- ☐ DN approval queue - approve same-day if delivery date is reasonable
- ☐ Vendor Returns - chase open returns > 7 days old

End of day:

- ☐ Confirm every PO assigned today shows in the receiving Warehouse's notification log
- ☐ T-1 dashboard (Admin Portal -> Logistics Oversight -> DNs sub-tab) - pre-empt tomorrow's deliveries

§3.6 Logistics - Weekly

- ☐ Review the WHATSAPP_TRIGGERS toggles in config.py - adjust if noise complaints came in
- ☐ Generate Warehouse Throughput report - flag underperforming warehouses to Warehouse Lead
- ☐ Generate Force-Closures report - review the week's closures with Admin
- ☐ Vendor performance review - open returns + delivery delays per vendor

§3.7 Logistics - Monthly

- ☐ Vendor master cleanup - deactivate dormant vendors (>90 days no PO)
- ☐ PR-to-PO cycle-time analysis - is the chain getting faster or slower?
- ☐ Force-closure trend - too many means upstream PR quality is poor

§3.8 Warehouse User - Per Shift

Shift start:

- ☐ bell badge - clear any unread assignments / DN rejections
- ☐ Incoming Assignments tab - [ok] Acknowledge new assignments within 1 hour
- ☐ Review today's Expected_Delivery list - physically prepare receiving area

During shift (when goods arrive):

- ☐ Receive Goods -> record actual qty per line as physical inspection completes
- ☐ If short-receive vs PO qty - note the discrepancy in remarks; Logistics will be notified
- ☐ If quality issue at receive - DON'T add to stock; raise via Returns from Site or a vendor-return note to Logistics

Before DN preparation:

- ☐ Confirm RL and BL items go on SEPARATE DNs (system enforces, but plan physically too)
- ☐ Verify destination site for each DN
- ☐ Driver + vehicle details ready

Shift end:

- ☐ Outbound DNs -> confirm every DN drafted today is submitted to Logistics (no orphan drafts)
- ☐ Sign off - audit-logged

§3.9 Warehouse User - Weekly

- [] History tab - sanity-check completed DNs match physical movement records
- [] Open assignments older than 7 days - chase Logistics for closure or vendor follow-up
- [] RL/BL inventory walk - physically verify families are separated in the warehouse

§3.10 Warehouse User - Monthly

- [] Throughput review with Warehouse Lead - DNs prepared, on-time rate, rejection rate
- [] Vehicle / driver master refresh - update Vehicle_No / Driver_Name defaults

§3.11 Store Keeper - Each Delivery Day

- [] Entry Log -> Receipt Staging - the Incoming Delivery Notes expander shows pending DNs
- [] Physical truck arrives -> inspect, count, match against DN
- [] If everything matches -> [ok] Mark as Received on the DN card
- [] If discrepancy -> tell HOD before marking; HOD escalates to Logistics

§3.12 Store Keeper - Standard Daily (unchanged from v2.0)

See USER_MANUAL.md §4 for the Consumption Log / Receipt Staging / Return Items / Returnable Items / Stock Count / QR Label Request daily routines.

§3.13 Admin - Weekly

- [] Admin Portal -> Logistics Oversight -> review every sub-tab
- [] Force-closures sub-tab - any not flagged "agreed-by-stakeholders" -> ask Logistics
- [] Users -> review pending registrations (Logistics + Warehouse users always go through admin approval)
- [] Audit Logs -> filter by FORCE_CLOSE_* actions

§3.14 Admin - Monthly

- [] Generate all 3 procurement reports (PO Status, Warehouse Throughput, Force-Closures)
- [] Bell-noise audit - check app_notifications row count vs prior month; sudden spike means a process change is needed
- [] Vendor + Warehouse master review - deactivate stale entries

§3.15 Admin - Quarterly

- [] WHATSAPP_TRIGGERS tuning meeting - review with Logistics, adjust per-event toggles
- [] SOP review - update this document with any process changes
- [] Database backup integrity test - restore to a sandbox + run bug_check.py

§4. Decision Trees

§4.1 DN approval - Approve vs Reject vs Reschedule

DN in HOD approval
queue (or Logistics)

Inspect: qty, family,
lot, expiry, vehicle, driver

Everything OK	Date wrong but content is fine	Content wrong
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[ok] Approve	Reschedule (set new date)	[x] Reject (must include reason)
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Logistics decides within 2h business	Warehouse rebuilds DN; loop continues
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Approve when: Qty matches PR - Family correct - Lot has > 90d shelf life - Vehicle + driver populated - No HSE concern with this delivery date

Reschedule (don't reject) when: Content is fine but YOU can't receive on the scheduled date (site shutdown, manpower, weather). Reschedule is faster than reject + rebuild.

Reject when: Qty doesn't match PR - Wrong material - Lot expiry inadequate - Wrong destination site (misroute) - Driver/vehicle missing critical info

§4.2 Vendor return vs Stock adjustment

Discrepancy discovered

Was material RECEIVED already
via a DN this site approved?

YES (in receipts)	NO (still at warehouse)
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Is the issue with THIS DN (defective, wrong qty), or general stock count drift?	Warehouse handles internally - Return to vendor flow (Tab 8 Warehouse)
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This DN	General drift
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Vendor	Stock Adjustment
Return	(SK Entry Log ->
(any role)	Stock Count tab)

Use Vendor Return when: A specific DN brought defective / wrong qty / quality-issue material. Reopens the originating PO line so Logistics can chase vendor.

Use Stock Adjustment when: Cycle-count discovers shelf qty system qty for a general reason (damage in storage, miscount, expired-disposal, etc.). Doesn't touch the PO chain.

§4.3 30-day return window - Override or not?

SK / HOD wants to return
material to logistics

Received in last 30d?

YES

NO

Standard return flow
(SK Entry Log -> Return)

Override required:
tick "Override 30-day window"
+ write justification

HOD reviews - override
rows highlighted RED

Justified

Not justified

Approve

Reject
(use Stock Adjustment
with reason `damaged`
or `expired\_disposal`)

§4.4 Force-close decision

Logistics considering
force-closing PR/PO/line

Has stakeholder been
consulted (Site HOD)?

NO YES

STOP - call HOD first. Proceed to next gate.
Force-close is permanent.

Is the reason in
scope of force-close?

In scope Out of scope

Force-close Use alternative:
with full reason - Vendor cancellation -> close PR
 - Wrong qty on PO -> vendor return + new PO
 - Wrong material -> vendor return
 - Project cancelled -> close PR

In scope for force-close: Project cancellation - Vendor permanently unable to fulfill - PR raised in error and beyond email-recall - Item discontinued - Compliance violation

Out of scope (use other tool): Quantity / quality issues (use vendor return) - Date slippage (use reschedule) - Wrong destination site (rebuild DN, no force-close needed)

§5. Escalation Matrix

When a process step doesn't happen on time, here's who pings whom, when, and on which channel.

§5.1 PR -> PO escalation

Trigger	Wait time	Action	Channel
PR submitted to Logistics, no PO after 4 business hours	4h	Site HOD -> Logistics lead	WhatsApp
Still nothing after another 4h	8h	Site HOD -> Admin	App notification + email
Still nothing after 24h	24h	Admin overrides - assigns alternate Logistics user	Admin Portal -> Users

§5.2 PO -> Assignment escalation

Trigger	Wait time	Action	Channel
PO issued, no warehouse assignment after 8 business hours	8h	Site HOD -> Logistics (with PO number)	WhatsApp
Assignment created, Warehouse hasn't acknowledged in 24h	24h	Logistics -> Warehouse Lead	WhatsApp + email
Still no ack after 48h	48h	Warehouse Lead -> Admin	Admin escalation

§5.3 DN approval escalation

Trigger	Wait time	Action	Channel
DN sitting at pending_logistics for > 4 business hours	4h	Warehouse -> Logistics	WhatsApp
DN sitting at pending_hod for > 8 business hours	8h	Logistics -> Site HOD	WhatsApp
DN sitting at pending_sk for > 24 hours after expected arrival	24h	HOD -> SK + Warehouse	WhatsApp

§5.4 Reschedule escalation

Trigger	Wait time	Action	Channel
Reschedule pending > 2 business hours during work day	2h	Requester -> Logistics	WhatsApp
Pending > 24h	24h	Site HOD -> Admin	App + email

§5.5 Vendor return escalation

Trigger	Wait time	Action	Channel
Vendor return raised, no acknowledgement from vendor in 7 days	7d	Logistics -> Vendor directly (email)	Email
Still unresolved after 14d	14d	Logistics -> Site HOD + Admin	App + email
Resupply date passed without delivery	T+1d	Logistics -> Admin	Critical

§5.6 Force-closure escalation

Force-closures fire critical notifications to Admin + originating Site HOD instantly. There's no escalation needed - the action is already at maximum severity. Admin reviews weekly per §3.13.

§6. Recovery Procedures

§6.1 DN rejected at HOD - how Warehouse recovers

1. Bell shows rejection notification with HOD's reason
2. Warehouse user reads the reason from the notification card (full text in the related DN row in delivery_notes.rejection_reason)
3. Determine what changed:

- If qty wrong: prepare a new DN with corrected qty (the rejected DN's qty frees back into the available calculation immediately)
 - If material wrong: file an internal Stock Adjustment if you over-counted at receive, OR raise a vendor return if the wrong material was delivered to you
 - If date wrong: prepare new DN with the corrected date; HOD will approve in same cycle
4. Submit new DN -> Logistics -> HOD; loop continues

§6.2 Warehouse short-receives from vendor

1. Receive Goods tab - type the actual received qty (less than ordered)
2. System auto-flips po_items.line_status to partially_delivered
3. Add a note in the assignment notes describing the shortfall
4. Notify Logistics via WhatsApp with the PO number + shortfall qty
5. Logistics raises a Vendor Return with reason "Vendor under-shipped" (uses the same return flow, reopens the line for resupply)
6. New shipment from vendor -> Warehouse receives the remainder -> line flips to delivered

§6.3 Vendor return + resupply cycle

1. Issue surfaces (SK at site OR Warehouse at receive)
2. Determine raiser:
 - Issue at site after SK confirmed -> SK raises to HOD -> HOD raises to Warehouse via internal return -> Warehouse raises Vendor Return
 - Issue at warehouse before DN ship -> Warehouse raises Vendor Return directly
3. Logistics tracks return in Vendor Returns tab
4. Expected_Resupply date set (default today + 14)
5. T-2 / T-1 / T-0 reminders fire on Expected_Resupply
6. When resupply arrives: Warehouse re-receives -> line Delivered_Qty recalculates -> status moves forward

§6.4 HOD absent for > 24h with pending DN

1. Admin shadows HOD Portal (admin can access via the existing override)
2. Admin reviews the pending DN(s) in HOD Portal -> DN Approvals
3. Admin approves on behalf, leaves a note in the DN attesting to the cover
4. Audit log records DN_HOD_APPROVE with admin's username
5. HOD-on-return is notified to review post-fact

§6.5 Logistics absent - Admin override

1. Admin enters Logistics Portal via shadow access
2. Admin acts as Logistics for the day (approve PRs, issue POs if pre-arranged, approve reschedules)
3. Audit log records each action with admin's username (not the absent Logistics user)
4. Admin notifies the absent Logistics user via WhatsApp before signing off

§6.6 Force-closure made in error

There's no "undo force-close" button (yet - on the v3.0 backlog). Workaround:

1. Admin -> Master DB Editor -> relevant table (purchase_orders / pr_master)
2. Edit status from force_closed back to open (PO) or remove the logistics_status='force_closed' flag (PR)

3. Add a corresponding po_force_closures row with reason "Reversed in error - see audit"
4. The Site HOD's "Force-closures affecting me" tab will show both the original close + the reversal

Use sparingly - this is invasive. Better: communicate the reversal verbally and update via DB Editor with a paper trail.

§6.7 RL/BL DN was prepared as combined (system rejected, user confused)

1. UI shows: "Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."
2. Warehouse user reduces Ship Qty to 0 on either the RL lines OR the BL lines (not both)
3. Save DN draft - succeeds with only one family
4. Prepare second DN with the other family - same source PO, different family
5. Submit both DNs separately; both flow through approval independently

§7. Quick Reference Cards

§7.1 Site HOD - One-page summary

Daily must-dos: Bell badge clear - Pending Receipts approved - DN Approvals queue empty - EOD commit by 17:00 - Sign out clean.

Submit PR to Logistics: HOD Portal -> PRs -> Submit PR(s) to Logistics expander -> multi-select -> Submit.

See what's incoming: In-Transit -> Active in-transit (sorted by closest-to-SK).

Request reschedule: In-Transit -> click DN card -> Request reschedule (date defaults to ETA + 3 days).

Escalate stuck PR: WhatsApp Logistics if no PO after 4h business hours.

Force-closure questions: Check In-Transit -> Force-closures sub-tab; escalate to Admin if reason looks weak.

§7.2 Logistics - One-page summary

Hourly must-dos: Bell badge <= 30min - Incoming PRs queue <= 4h - Reschedules <= 2h.

Issue PO from PR: Incoming PRs -> drill in -> Use this PR to create a PO -> fill header -> save.

Issue PO from vendor PDF: Create PO -> PDF upload -> Extract -> Review -> Save.

Assign to warehouse: Assign -> pick PO + WH -> confirm items -> Expected Delivery -> Assign.

Approve DN date: Outbound DNs queue (Tab 4 visible on Warehouse Portal - Logistics actions through DN approval queue) -> [ok] Approve.

Force-close: Always call Site HOD first. Then Force-Close with detailed reason.

§7.3 Warehouse User - One-page summary

Per shift: Bell badge clear - Incoming Assignments acked within 1h - Outbound DNs all submitted (no orphan drafts).

Receive goods: Receive Goods -> assignment -> type Receive Now per line -> Record receipt.

Prepare DN: Prepare DN -> pick PO + site -> Ship Qty per line -> header -> Save -> Submit.

Strict rule: ONE family (RL or BL) per DN. System rejects mixed.

Returns from site: Returns from Site -> DN -> return qty + reason -> Raise to vendor.

§7.4 Store Keeper - One-page summary

On delivery day: Entry Log -> Receipt Staging -> Incoming DN expander -> inspect -> [ok] Mark as Received.

On every day: Consumption Log -> record issues; Receipt Staging for direct deliveries; Stock Count for cycle

counts; QR Label Request for new labels.

Don't: Mark received without physical inspection. Mix Returnable (tool loans) with Return Items (real returns).

Smart Scan - when CV detection is wrong: the green "Auto-filled" card can be edited before submit. If a yellow "Top candidates" radio appears, pick the right one or override the tool name in the manual form. If no candidates show (or no active model), just type it manually - borrower fields stay pre-filled from the badge scan. The submitted loan records which path (CV vs manual) was used so adoption telemetry stays honest. Full workflow: USER_MANUAL.md §4.5.0. Trainer guide: docs/cv_training_guide.md.

§7.5 Admin - One-page summary

Weekly: Logistics Oversight scan; force-closures review; user registrations approve.

Monthly: 3 procurement reports (PO Status / Warehouse Throughput / Force-Closures); bell-noise audit.

Quarterly: SOP review; backup integrity test.

Override: Use Master DB Editor with a paper trail; document every override in system_audit_log via the affected helper.

§8. Glossary

For complete vocabulary (FEFO, EOD, PR, DN, OCR, WAL, etc.) see USER_MANUAL.md §11.5. Procurement-chain-specific terms only here:

Term	Meaning
Procurement chain	The in-app SQL-driven workflow from Site PR -> Logistics PO -> Warehouse DN -> Site SK receipt. Replaces (optionally) the legacy email/PDF flow.
DN state machine	The seven-step lifecycle of a Delivery Note (draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received), with rejected as terminal from any pending state.
RL/BL strict separation	Hard rule that Rubber Lining and Brick Lining never share a PO group, DN, or warehouse aggregation. Enforced at three layers.
Warehouse-blind pricing	Hard rule that warehouse_user role never sees Unit_Price, Total_Price, or monetary header fields on any PO. Three enforcement layers.
Logistics shadow	Admin's read+write access to the Logistics Portal as if they were a Logistics user. Used for absence cover.
Warehouse shadow	Admin's access to the Warehouse Portal with a sidebar warehouse picker. Used for absence cover or audit.
Force-closure	Logistics' one-way close of a PR / PO / line with mandatory reason. Notifies Admin + Site HOD with critical severity.
T-2 / T-1 / T-0 reminders	Daily sweep job firing notifications 2 days before, 1 day before, and on the expected delivery date. Idempotent.
App notification	A row in app_notifications, surfaced via the sidebar bell. Always fires; in-app only.
WhatsApp trigger toggle	config.WHATSAPP_TRIGGERS[event_key] = True/False to per-event control WhatsApp emission. In-app is unaffected.

Term	Meaning
Material Estimator (SME)	HOD/Admin planning portal for Rubber/Brick-Lining material. Read-only projection over the ledger; Available_Qty is computed, never stored. SME inventory is isolated from the ERP inventory table.
Man-Hours	HOD/Admin labor-tracking module. Employee roster + daily timesheets (Excel upload or manual grid) + team-SQM distribution + a man-hour estimator + an estimate-vs-actual variance dashboard. Writes only mh_* tables; never touches inventory or the EOD ledger.
OWN vs Supply	Labor type on the Man-Hours roster: OWN = your own (GI) staff; Supply = subcontractor labor (e.g. DMC).
Normal vs OT hours	Man-Hours computes Total = (Out In) break, Normal = min(Total, 8h), OT = max(0, Total - 8h). The source file's own hour columns are ignored.
SQM distribution	How a day's team SQM is split to each worker's Allocated_SQM: even (equal per worker) or by-hours (pro-rata on hours).

§9. Change Log

Version	Date	Author	Changes
1.0	2026-06	Initial release	First publication alongside Hub v3.0 procurement chain. Covers Site HOD, Logistics, Warehouse User, SK, Admin. Includes RACI, daily/weekly/monthly cadences, 4 decision trees, 6-tier escalation matrix, 7 recovery procedures, 5 quick-reference cards.

Version	Date	Author	Changes
1.1	2026-06-28	Update	Added §3.3a Man-Hours & Labor Tracking and §3.3b Material Estimator HOD cadences; glossary entries for both modules (SME isolation, OWN/Supply, Normal/OT, SQM distribution). These HOD/Admin-only modules are site-scoped and isolated from the inventory ledger.

End of SOP. Review quarterly. Update on any role change, vendor onboarding, escalation-time tweak, or DN-state-machine alteration. Companion: USER_MANUAL.md (UI reference) - handoff.md (technical architecture).